BeautyCo — Association Rules: Answers to Questions 3–5

Student ID: 6710503739 | 1,000 transactions, 14 products | FP-Growth (support ≥ 0.2), rules (confidence ≥ 0.7)

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1. EDA

What does each row represent?
One customer's purchase transaction (one basket). Each column is a product: `1` = purchased, `0` = not purchased. 1,000 baskets, 14 products, no missing values, 4.39 items per basket on average.

Which products are most frequently purchased together?

Product A	Product B	Support	Lift
Base_Makeup	Glossy_Lips	0.356	1.36
Eye_Enhancer	Eye_Palette	0.321	2.36
Coverage_Base	Eye_Liner	0.297	1.47
Base_Makeup	Eye_Liner	0.238	0.97
Coverage_Base	Base_Makeup	0.231	0.98

`Base_Makeup + Glossy_Lips` is the largest by volume (0.356), but its lift is only 1.36. `Eye_Enhancer + Eye_Palette` (0.321, lift 2.36) is the genuinely related pair.

Are there any unexpected product combinations?

- `Nail_Art + Makeup_Tools` — **lift 3.57**, the highest in the dataset, but support 0.149 is below the 0.2 threshold, so FP-Growth never reports it.
- `Lip_Liner + Base_Makeup` — **lift 0.69**, a negative association: these two are avoided together, even though Base_Makeup is the best seller.

2. Association Rule Modeling

How many frequent itemsets were discovered?
23 — 11 single items and 12 pairs. Nothing of size 3 or larger reaches 0.2 support.

Identify the maximal frequent itemsets.
16 of 23 — all 12 pairs, plus 4 single items that pair with nothing at 0.2 support.

Maximal pair	Support	Maximal pair	Support
Base_Makeup, Glossy_Lips	0.356	Eye_Liner, Glossy_Lips	0.227
Eye_Enhancer, Eye_Palette	0.321	Cheek_Color, Coverage_Base	0.220
Coverage_Base, Eye_Liner	0.297	Coverage_Base, Glossy_Lips	0.218
Base_Makeup, Eye_Liner	0.238	Base_Makeup, Eye_Palette	0.211
Base_Makeup, Coverage_Base	0.231	Coverage_Base, Eye_Enhancer	0.204
		Coverage_Base, Eye_Palette	0.201
		Eye_Palette, Glossy_Lips	0.201
Maximal singleton	**Support**	**Maximal singleton**	**Support**
Lip_Color	0.322	Contouring	0.279
Nail_Art	0.280	Lip_Liner	0.234

3. Deployment and Interpretation

Rule with the highest confidence
Only 3 rules reach confidence 0.7. The highest is:

IF `Eye_Enhancer` (mascara) THEN `Eye_Palette` (eyeshadow palette)

Metric	Value	Calculation
Support	0.321	$321/1{,}000$ transactions
Confidence	0.899	$321/357$ mascara buyers
Lift	2.36	$0.899/0.381$

Meaning
32.1% of all baskets contain both products. Of the customers who buy mascara, 89.9% also buy an eyeshadow palette — against 38.1% for customers in general, so mascara buyers are 2.36$\times$ more likely than average to buy a palette. Lift > 1 confirms a real association, not coincidence.

Marketing strategy

- **Product bundle:** sell `Eye_Enhancer + Eye_Palette` together as an "Eye Look Set". One basket in three already contains both.
- **TikTok content:** eye-makeup tutorials that apply the palette and then the mascara, so one clip sells both products.
- **Promotional campaign:** recommend the palette whenever mascara is added to the cart, and retarget the 36 mascara buyers who did not buy one.